

DTMamba : Dual Twin Mamba for Time Series Forecasting

Zexue Wu, Yifeng Gong, Aoqian Zhang*, and Boyang Li

Abstract: Long-term Time Series Forecasting (LTSF) has always been an important task where models need to effectively capture hidden patterns in the time series in order to make accurate predictions about future states. The State-Of-The-Art (SOTA) models in this area are mostly based on Transformers, but the prediction accuracy still needs to be improved. Recently, Mamba has emerged as a promising approach for modeling sequential data, especially for autoregressive tasks with long sequences. Therefore, in this paper, we propose an LTSF model called DTMamba. DTMamba utilizes innovative dual twin Mamba blocks to extract long-term dependencies in time series, while also incorporating residual network to enhance the overall predictive capability. We perform experiments with 8 publicly available datasets and compare DTMamba with 11 SOTA models. The experimental results show that DTMamba outperforms the SOTA models in terms of performance. In particular, our proposal has obvious advantages when dealing with low-dimensional data and predicting longer time series.

Key words: time series forecasting; Mamba; multivariate time series

1 Introduction

Long-term Time Series Forecasting (LTSF) is of paramount importance as it enables accurate predictions of future trends based on historical data. Accurate time series forecasting is of great use in various fields. For example, Ref. [1] achieves accurate prediction of the Shanghai Composite Index, and Ref. [2] precisely predicts both long-term and short-term demands in the supply chain. Both of the aforementioned examples have brought significant economic value. In the field of healthcare and wellness, Ref. [3] predicts Key Performance Indicators (KPIs) in healthcare, which can facilitate informed planning and improve the quality of medical services. Reference [4] demonstrates excellent predictive capabilities for specific numerical values in weather, such as minimum

temperature, maximum temperature, wind speed, and others.

In the past, some traditional methods, such as statistical-based method^[5], have shown limited performance in capturing the complex and non-linear relationships among data. With the advent of Recurrent Neural Networks (RNNs)^[6] and Convolutional Neural Networks (CNNs)^[7], RNN-based^[8] and CNN-based models^[9] have outperformed traditional methods in terms of performance and robustness. On the other hand, Transformer-based models^[10, 11] have reached new heights. They have successfully utilized the self-attention mechanisms of Transformers to capture the long-term dependencies in time series data and achieve State-Of-The-Art (SOTA) performance. However, Transformer-based models still have limitations in modeling long sequences, which leads to suboptimal prediction performance. In recent years, some researchers have found that simple linear models^[12, 13] can achieve comparable performance to Transformer-based models while exhibiting lower model complexity.

Recently, Mamba^[14] has emerged as an innovative approach to modeling sequences. By using the

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framework of State Space Models (SSM)^[15], Mamba achieves a fusion of the sequential processing capability of RNN and the global information processing capability of CNN. With a selective mechanism within the SSM framework, it can focus on the essential information while filtering out irrelevant details, thus enables efficient training and inference processes. In addition, Mamba has shown considerable potential in various areas, including graph^[16], images^[17], and language^[18].

In this paper, we propose a novel model based on Mamba, called Dual Twin Mamba (DTMamba), to further capture the long-term dependencies in time series and improve the capability of modeling long sequences. After normalization and processing by the channel independence technique^[11], the input data are then fed into two sequentially connected Twin Mamba (TMamba) blocks (see Section 3.3), which aim to extract data correlations in long sequences. Finally, a projection module, a reverse channel independence module, and a reverse normalization module are applied sequentially to generate the prediction results.

In summary, we make the following three contributions in this paper:

- We propose a novel Mamba-based model for long-term multivariate time series forecasting namely DTMamba, which uses channel independence to reduce overfitting and employs two sequentially connected TMamba blocks to better learn hidden information.
- We propose the TMamba block to extract long-term dependencies in time series through two parallel Mamba blocks and residuals. We also integrate a linear layer and a dropout layer to improve the performance of the TMamba block.
- We evaluate the performance of DTMamba and show its superiority over other SOTA models. We validate the effectiveness of the modules in the architecture via ablation studies, and the robustness to hyper-parameters through sensitivity tests. In particular, the proposed DTMamba shows clearer advantages with low-dimensional data and the prediction of longer time periods.

Table 1 lists the notations frequently used in this paper.

2 Related Work

2.1 Transformer-based methods

Autoformer^[19] innovatively adopts a decomposition

Table 1 Notation.

Symbol	Description
X	Historical multivariate time series
$\hat{X}$	Predicted multivariate time series
X^0	Normalization result of X
X^I	Channel independence result of X^0 or the result of adding $X^{M1,i}$ and $X^{M2,i}$
TMamba_ i	The i -th TMamba block in DTMamba, $i \in \{1, 2\}$
Linear_ i	Linear layer in TMamba_ i , $i \in \{1, 2\}$
$X^{L,i}$	Result of i -th linear layer
R^i	Residual of TMamba_ i
$X^{D,i}$	Dropout result of TMamba_ i
$X^{M1,i}$	the first Mamba result of TMamba_ i
$X^{M2,i}$	the second Mamba result of TMamba_ i
X^A	Result of adding DTMamba's results, R^1 and R^2 , i.e., residual connection
X^P	Projection result

architecture integrated with a special auto-correlation mechanism, deviating from the traditional series decomposition pre-processing norm and revolutionizing it as a fundamental building block within deep models. Hence, it is capable of progressively decomposing complex time series to extract their inherent features.

FEDformer^[20] integrates the Transformer architecture with seasonal-trend decomposition approach, introducing a novel approach known as frequency-enhanced Transformer. This innovative model combines the power of Transformer-based deep learning with the sophisticated analysis of seasonal patterns and trends, enabling enhanced frequency-aware modeling and prediction capabilities.

Stationary^[21] is dedicated to balancing the predictive capability of models for time series data and their model capacity by incorporating two interconnected modules: series stationarization and de-stationary attention. These modules collaboratively contribute to the overall functionality of the model, enabling effective handling of non-stationary time series data while preserving predictive accuracy. Crossformer^[22] introduces a Transformer-based model that leverages DSW embedding and Two-Stage Attention (TSA) layers to construct a Hierarchical Encoder-Decoder (HED) architecture, facilitating accurate predictions. This innovative approach enhances the model's ability to capture intricate relationships and dependencies within the data, resulting in improved prediction performance. However, in TSA layer, Crossformer

establishes a straightforward interconnection among dimensions, which may potentially cause noise in datasets with high dimensionality. PatchTST^[11] introduces a novel approach for time series analysis. It segments time series into sub-series level patches, which are used as input tokens in the Transformer model. Additionally, PatchTST leverages channel independence to enhance prediction accuracy. This unique methodology significantly improves the model's ability to handle time series data and achieve accurate predictions. However, PatchTST treats channels as completely independent without considering potential correlations between them, which may lead to a decrease in prediction accuracy. iTransformer^[10] introduces a transformative approach by applying attention and feed-forward networks to the inverted dimensions within the traditional Transformer architecture. This innovative modification enables the model to effectively capture and leverage patterns and dependencies across the inverted dimensions, leading to improved performance and enhanced capabilities. Although Transformer-based methods are considered mainstream algorithms, Transformer-based methods are still inadequate in terms of prediction performance.

2.2 Other methods

Dlinear^[12], TiDE^[13], and Rlinear^[23] are all linear-based approaches. Dlinear recognizes the inherent limitation of the self-attention mechanism with permutation-invariance, which results in the loss of temporal information. As a solution, Dlinear adopts a set of simple single-layer linear models to effectively model time series data. By leveraging these straightforward yet effective models, Dlinear mitigates the loss of temporal information and improves its ability to capture and analyze time-dependent patterns. However, Dlinear is limited by model capacity. TiDE just employs a straightforward encoder-decoder architecture based on MultiLayer Perception (MLP) to conduct long-term time series forecasting. The simplicity of this architecture is notable. Rlinear deviates from intricate architectures and achieves competitive forecasting performance through the utilization of simple linear layers, which is similar to TiDE. The performance of Rlinear is often suboptimal on datasets with significant seasonality. These linear-based approaches exhibit higher time and memory efficiency compared to the Transformer-based methods. However, in scenarios exhibiting significant

fluctuations and lacking periodicity and smoothness, linear-based models relying solely on temporal patterns observed in the past cannot perform satisfactorily^[24]. SCINet^[25] introduces a recursive architecture that involves downsampling, convolving, and interacting operations. SCINet employs multiple convolutional layers to capture distinct yet important temporal characteristics from downsampled subsequences or features. TimesNet^[26] employs a modularized structure to decompose intricate temporal patterns into discrete periods. By converting the initial unidimensional time series into a bidimensional space, TimesNet accomplishes integrated modeling of both intra-periodic and inter-periodic patterns. Both the Temporal Convolutional Network (TCN)-based SCINet and TimesNet show limited modeling capabilities for long-term dependencies.

2.3 Applications of Mamba

Mamba is a novel architecture that stands out from all existing frameworks and overcomes the limitations of the previously mentioned methods. There are currently variants based on Mamba, such as U-Mamba^[17], which has been applied to biomedical image segmentation and has achieved SOTA results on related tasks. Another variant is Vim^[27], which enables efficient learning of visual representations and shows significantly improved inference speed and memory utilization compared to previous Transformer-based models. In addition, VideoMamba^[28] is designed for efficient video comprehension and shows advantages in long video comprehension tasks.

3 DTMamba

In this section, we will first define the problem of time series data forecasting. Then, we will introduce the proposed architecture of DTMamba model and the novel TMamba block used to address this problem.

3.1 Problem statement

Definition 1 (Multivariate time series) Consider a multivariate time series of L observations, $X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_L\}$. The i -th observation (data point) $\mathbf{x}_i \in \mathbf{R}^N$ consists of N dimensions $\{\mathbf{x}_i^1, \mathbf{x}_i^2, \dots, \mathbf{x}_i^N\}$ and is observed at time t_i .

Problem 1 Given a historical multivariate time series $X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\} \in \mathbf{R}^{T \times N}$, the long-term multivariate time series forecasting problem is to predict $\hat{X} = \{\hat{\mathbf{x}}_{T+1}, \hat{\mathbf{x}}_{T+2}, \dots, \hat{\mathbf{x}}_{T+S}\} \in \mathbf{R}^{S \times N}$, where $T \leq S$.

3.2 Architecture

In this section, we provide a detailed introduction to the architecture of the proposed DTMamba, shown in Fig. 1.

Before the input multivariate time series data $X \in \mathbf{R}^{T \times N}$ is fed into TMamba block, we first normalize it into $X^0 = \text{RevIN}(X) = \{x_1^0, x_2^0, \dots, x_T^0\} \in \mathbf{R}^{T \times N}$. Here, $\text{RevIN}(\cdot)$ stands for the normalization method $\text{RevIN}^{[29]}$ and $\text{RevIN}^{-1}(\cdot)$ stands for the reverse normalization method.

Channel independence can prevent model overfitting in time series forecasting^[11]. Therefore, we perform such a technique on the normalized data to ensure that each dimension is processed independently in the following computations. $\text{ChannelIndependence}(\cdot)$ will reshape $X^0 : (T, N)$ (lookback length and dimension number) into $X^1 : (N, T)$.

After the above data processing, the time series data pass through the two sequentially connected TMamba blocks. The details of TMamba Block can be found in Section 3.3. The projection layer then uses the hidden information extracted from the TMamba blocks and the residual terms to predict the subsequent time series data.

Since we need to obtain the predicted results $\hat{X} \in \mathbf{R}^{S \times N}$ at the end, we need to perform the inverse channel independence operation to reshape the data and finally denormalize it.

Algorithm 1 presents the forecasting procedure of the DTMamba model in detail. Our model uses two sequentially connected TMamba blocks corresponding to Line 3 in Algorithm 1, where $i \in \{1, 2\}$.

3.3 Twin Mamba

As shown in Fig. 1, the TMamba block consists of a linear layer. Its output serves not only as a residual term for subsequent calculations, but also as input for the dropout layer. The output of the dropout layer is then used as input for the twin Mambas (with the same model hyperparameter settings). The TMamba block outputs the residuals and hidden information learned from this block.

3.3.1 Linear layer

Linear models have unique advantages in predicting time series^[30], so we use a linear layer to transform the dimension. In addition, we transform the entire lookback length of the time series to obtain the global feature representation of the time series.

As shown in Fig. 1, DTMamba consists of two TMamba blocks in total. Hence, the linear layer transforms the input data into different dimensions, namely n_1 and n_2 . So we will transform the X^1 into $X^{L,i} : (N, n_i)$. The sensitivity of n_1 and n_2 is discussed in Section 4.3.5. In the following discussion, we will refer to the linear layer in the first TMamba block as Linear_1 and in the second block as Linear_2 .

3.3.2 Residual

The residual connection was first proposed by ResNet^[31], and so far it has been shown in numerous studies to prevent overfitting, provide more stable training and achieve better fitting^[32]. Therefore, we add a residual term after the linear layer, and to fulfill the dimensionality requirements of the second TMamba block n_2 , we use an Full Convolutional (FC) layer to transform the dimensionality of the residual. Similarly,

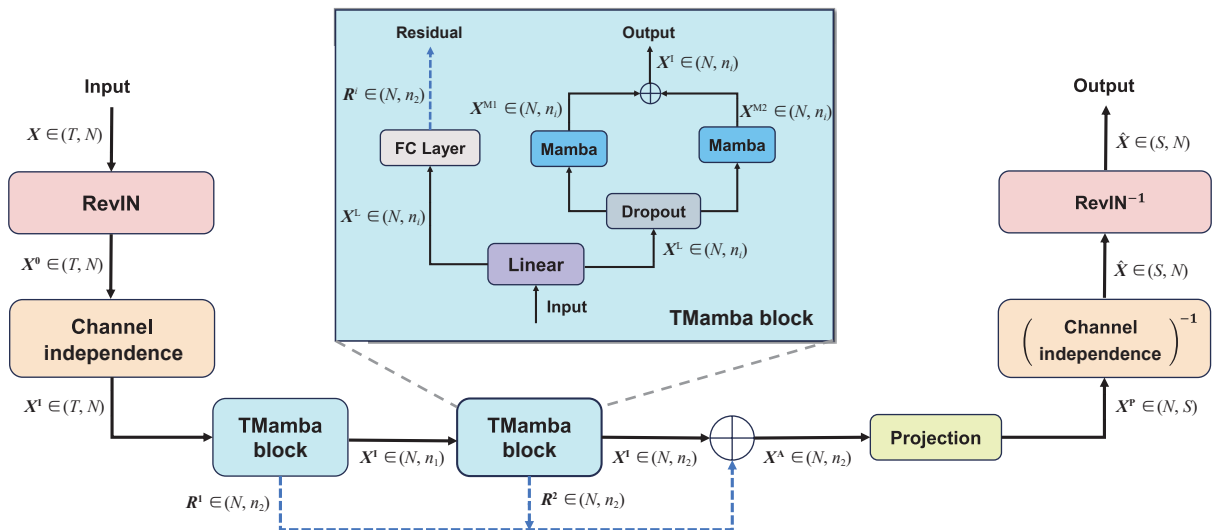


Fig. 1 Architecture of DTMamba.

Algorithm 1 Forecasting procedure of DTMamba

Input: $X = [x_1, x_2, \dots, x_T]: (T, N)$
Output: $\hat{X} = [\hat{x}_{T+1}, \hat{x}_{T+2}, \dots, \hat{x}_{T+S}]: (S, N)$

- 1 $X^0: (T, N) \leftarrow \text{RevIN}(X)$ {Normalization};
- 2 $X^1: (N, T) \leftarrow \text{ChannelIndependence}(X^0)$;
- 3 **for** TMamba $_i$ in TMambaBlocks **do**
- 4 $X^{L,i}: (N, n_i) \leftarrow \text{Linear}_i(X^1)$;
- 5 $R^i: (N, n_2) \leftarrow \text{FC}(X^{L,i})$ {Residual};
- 6 $X^{D,i}: (N, n_i) \leftarrow \text{Dropout}(X^{L,i})$;
- 7 $X^{M1,i}: (N, n_i) \leftarrow \text{Mamba}(X^{D,i})$;
- 8 $X^{M2,i}: (N, n_i) \leftarrow \text{Mamba}(X^{D,i})$;
- 9 $X^i: (N, n_i) \leftarrow X^{M1,i} + X^{M2,i}$
- 10 $X^A: (N, n_2) \leftarrow X^1 + R^1 + R^2$ {Residual connection};
- 11 $X^P: (N, S) \leftarrow \text{Projection}(X^A)$ {Projection};
- 12 $\hat{X}: (S, N) \leftarrow \text{ChannelIndependence}^{-1}(X^P)$;
- 13 $\hat{X}: (S, N) \leftarrow \text{RevIN}^{-1}(\hat{X})$;
- 14 **return** $\hat{X}$

we will denote the residual in the first as R^1 and in the second block as R^2 .

3.3.3 Mamba

Mamba is primarily built upon Structural State Space Sequence (S4) model, a recent State Space Model (SSM) architecture^[14]. At a high level, S4 acquires the ability to establish a mapping between an input $x(t)$ and an output $y(t)$ by utilizing an intermediary state $h(t)$. Here, x , y , and h are functions of time t , as SSMs are specifically designed to effectively handle continuous data types, such as audio, sensor data, and images. S4 establishes the interrelationships among these variables using three continuous parameter matrices: A , B , and C . These are interconnected through the following two equations:

$$h'(t) = A \cdot h(t) + B \cdot x(t) \quad (1)$$

$$y(t) = C \cdot h(t) \quad (2)$$

In practice, we always deal with discrete data, such as text. This requires us to discretize the SSM, transforming our continuous parameters A , B , and C into discrete parameters $\bar{A}$, $\bar{B}$, and $\bar{C}$ by using a special fourth parameter Δ ,

$$\bar{A} = \exp(\Delta \cdot A) \quad (3)$$

$$\bar{B} = (\Delta \cdot A)^{-1} (\exp(\Delta \cdot A) - I) \Delta \cdot B \quad (4)$$

Once discretized, we can instead represent the SSM through these two equations:

$$h_t = \bar{A} \cdot h_{t-1} + \bar{B} \cdot x_t \quad (5)$$

$$y_t = C \cdot h_t \quad (6)$$

These equations form a recurrence, similar to RNN^[33]. At each step t , we combine the hidden state from the previous timestep h_{t-1} with the current input x_t to create the new hidden state h_t .

From Eq. (5), it can be observed that h_t can only access information from the previous and current time steps. This token mixture is called a causal pattern, which can be expressed as

$$y_t = f(x_1, x_2, \dots, x_t) \quad (7)$$

where x_t and y_t represent the input and output of the t -th token, respectively. Due to its causal nature, this pattern is highly suitable for autoregressive tasks^[34]. Time series data forecasting is a typical autoregressive task, and Mamba is highly suitable for this task.

We use two parallel Mamba in the TMamba block. The parallel connection of two Mamba models can provide better expressive power. Each Mamba model has its own parameters and hidden states, so they can independently learn and represent features of time series data, better capturing the complex relationships and patterns in the data.

In addition, two sequentially connected TMamba blocks are used in this work, but the dimensions of the linear layers in the two TMamba blocks are different, with the first being 256 and the second being 128. The first TMamba block learns higher dimensional temporal features, while the second learns lower dimensional temporal features.

4 Experiment

In this section, we conduct various time series forecasting experiments on widely recognized benchmark datasets to comprehensively evaluate the performance of our proposed model against recognized forecasting models.

4.1 Settings

All experiments are conducted using two NVIDIA 2X3090 GPUs (24 GB each) within the PyTorch framework^[35], with the Ubuntu 20.04 operating system. The model is optimized using the ADAM algorithm^[36] with $L2$ regularization. Due to the limitations of GPU memory, the batch size is adjusted based on different datasets. The training for each dataset is set to 10 epochs.

4.1.1 Datasets

We conduct experiments with 8 real-world datasets widely used in the LTSF task, including Weather,

Traffic, Electricity, Exchange, and four ETTs (ETTh1, ETTh2, ETTm1, and ETTm2). The detailed descriptions of the datasets can be found in Table 2.

4.1.2 Metrics

The evaluation metrics used are Mean Squared Error (MSE)^[37] and Mean Absolute Error (MAE)^[38],

$$\text{MSE} = \frac{1}{n} \sum_{t=1}^n |X_t - \hat{X}_t|^2 \quad (8)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |X_t - \hat{X}_t| \quad (9)$$

Here $\hat{X}_t$ represents the predicted value at time t . Lower values of MSE and MAE indicate more accurate prediction results.

4.1.3 Baselines

We compare our model DTMamba with 11 SOTA models, including (1) Transformer-based methods: Autoformer^[19], FEDformer^[20], Stationary^[21], Crossformer^[22], PatchTST^[11], and iTransformer^[10]; (2) Linear-based methods: DLinear^[12], TiDE^[13], and RLinear^[23]; and (3) TCN-based methods: SCINet^[25] and TimesNet^[26].

4.1.4 Hyperparameters

All experimental results of the proposed model are obtained under fixed hyperparameters. Specifically, the settings are as follows: Dropout = 0.05 (except Electricity, where it is set to 0.5), Linear_1 = 256, Linear_2 = 128, and the three parameters of Mamba: dimension expansion factor (expand) = 1; state expansion factor (d_state) = 256; local convolutional width (d_conv) = 2.

4.2 Long-term forecasting

We conduct comprehensive time series forecasting experiments on various datasets.

4.2.1 Comparison among our proposals

Table 3 presents comprehensive prediction results, with

the lookback length and forecast length settings consistent with mainstream LTSF work. Overall, our model demonstrates the best experimental results. Moreover, iTransformer has the second best performance and shows clear advantages in the Traffic and Electricity datasets. It is evident that our approach performs well on datasets with lower dimensions and ranks just below iTransformer on datasets with higher dimensions, i.e., Traffic and Electricity.

To further validate the superiority of DTMamba in lower dimensions, we perform prediction experiments on the Electricity dataset (where the performance is worse than iTransformer). We choose $n \in \{1, 5, 10, 100\}$ dimensions and run experiments with a forecast length of $S \in \{96, 192, 336, 720\}$, and then average the results. As can be seen in Fig. 2, in the lower dimensions, such as 1, 5, and 10, both the MSE and MAE values for DTMamba are lower than those of iTransformer. DTMamba has better performance. However, if the number of dimensions continues to increase to a higher value such as 100, the performance of DTMamba may slightly lag behind iTransformer. It is pointed out that there are often “distribution drifts” in real data^[39], which means that there are significant differences between future sequences and historical sequences. Due to the fact that DTMamba consists of four Mamba, a larger model capacity may be more prone to overfitting for complex data with higher dimensions, resulting in poorer final prediction results.

To provide a more intuitive representation of the predictive ability of DTMamba, we visualize the prediction results of DTMamba and the second ranked iTransformer. We select four datasets, namely ETTh1, ETTm1, Exchange, and Weather, with both the lookback length and forecast length being set to 96. For each dataset, we select data from one dimension. As shown in Fig. 3, the orange line represents the ground truth and the blue line represents the prediction results. It is obvious that the prediction results of DTMamba are closer to the ground truth, even though the performance of all models is sometimes not satisfactory.

4.2.2 Varying lookback length T

In addition to the mainstream lookback length of 96, we also perform experiments with our DTMamba with lookback lengths of 192, 336, and 720. As shown in Table 4, the performance of DTMamba does not necessarily improve with an increase in lookback length for different prediction lengths. Similar results

Table 2 Summary of datasets.

Dataset ^[19]	Dimension	Number of time points	Frequency	Information
Weather	21	52 696	Per 10 min	Weather
Traffic	862	17 544	Hourly	Transportation
Electricity	321	26 304	Hourly	Electricity
Exchange	8	7588	Daily	Economy
ETTh1	7	17 420	Hourly	Electricity
ETTh2	7	17 420	Hourly	Electricity
ETTm1	7	69 680	Per 15 min	Electricity
ETTm2	7	69 680	Per 15 min	Electricity

Table 3 Prediction results of DTMamba and the baseline models for various time series forecasting. The lookback length T is fixed at 96, and the forecast length $S \in \{96, 192, 336, 720\}$. The best result is highlighted in bold black (“1st count” denotes the number of the best results), and the second-best result is underlined. “Avg” represent the average result across all forecast lengths.

Dataset	S	DTMamba		iTransformer		RLinear		PatchTST		Crossformer		TiDE		TimesNet		DLinear		SCINet		FEDformer		Stationary		Autoformer	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Weather	96	0.171	0.218	0.174	0.214	0.192	0.232	0.177	0.218	0.158	0.230	0.202	0.261	0.172	0.220	0.196	0.255	0.221	0.306	0.217	0.296	0.173	0.223	0.266	0.336
	192	0.220	<u>0.257</u>	0.221	0.254	0.240	0.271	0.225	0.259	0.206	0.277	0.242	0.298	<u>0.219</u>	0.261	0.237	0.296	0.261	0.340	0.276	0.336	0.245	0.285	0.307	0.367
	336	0.274	0.296	0.278	0.296	0.292	0.307	0.278	<u>0.297</u>	0.272	0.335	0.287	0.335	0.280	0.306	0.283	0.335	0.309	0.378	0.339	0.380	0.321	0.338	0.359	0.395
	720	0.346	0.344	0.358	<u>0.347</u>	0.364	0.353	0.354	{0.348}	0.398	0.418	0.351	0.386	0.365	0.359	0.345	0.381	0.377	0.427	0.403	0.428	0.414	0.410	0.419	0.428
	Avg	0.253	<u>0.279</u>	<u>0.258</u>	0.278	0.272	0.291	0.259	0.281	0.259	0.315	0.271	0.320	0.259	0.287	0.265	0.317	0.292	0.363	0.309	0.360	0.288	0.314	0.338	0.382
Traffic	96	0.487	0.317	0.395	0.268	0.649	0.389	0.462	0.295	0.522	<u>0.290</u>	0.805	0.493	0.593	0.321	0.650	0.396	0.788	0.499	0.587	0.366	0.612	0.338	0.613	0.388
	192	0.498	<u>0.325</u>	0.417	0.276	0.601	0.366	0.466	0.296	0.530	0.293	0.756	0.474	0.617	0.336	0.598	0.370	0.789	0.505	0.604	0.373	0.613	0.340	0.616	0.382
	336	0.511	0.334	0.433	0.283	0.609	0.369	0.482	0.304	0.558	<u>0.305</u>	0.762	0.477	0.629	0.336	0.605	0.373	0.797	0.508	0.621	0.383	0.618	0.328	0.622	0.337
	720	0.533	<u>0.326</u>	0.467	0.302	0.647	0.387	0.514	0.322	0.589	0.328	0.719	0.449	0.640	0.350	0.645	0.394	0.841	0.523	0.626	0.382	0.653	0.355	0.660	0.408
	Avg	0.507	0.326	0.428	0.282	0.626	0.378	<u>0.481</u>	<u>0.304</u>	0.550	<u>0.304</u>	0.760	0.473	0.620	0.336	0.625	0.383	0.804	0.509	0.610	0.376	0.624	0.340	0.628	0.379
Exchange	96	0.083	0.201	<u>0.086</u>	0.206	0.093	0.217	0.088	<u>0.205</u>	0.256	0.367	0.094	0.218	0.107	0.234	0.088	0.218	0.267	0.396	0.148	0.278	0.111	0.237	0.197	0.323
	192	0.173	0.295	0.177	<u>0.299</u>	0.184	0.307	<u>0.176</u>	<u>0.299</u>	0.470	0.509	0.184	0.307	0.226	0.344	<u>0.176</u>	0.315	0.351	0.459	0.271	0.315	0.219	0.335	0.300	0.369
	336	0.346	0.427	0.331	<u>0.417</u>	0.351	0.432	0.301	0.397	1.268	0.883	0.349	0.431	0.367	0.448	<u>0.313</u>	0.427	1.324	0.853	0.460	0.427	0.421	0.476	0.509	0.524
	720	0.868	0.698	<u>0.847</u>	0.691	0.886	0.714	0.901	0.714	1.767	1.068	0.852	0.698	0.964	0.746	0.839	<u>0.695</u>	1.058	0.797	1.195	<u>0.695</u>	1.092	0.769	1.447	0.941
	Avg	0.368	0.405	<u>0.360</u>	0.403	0.378	0.417	0.367	<u>0.404</u>	0.940	0.707	0.370	0.413	0.416	0.443	0.354	0.414	0.750	0.626	0.519	0.429	0.461	0.454	0.613	0.539
Electricity	96	0.166	<u>0.256</u>	0.148	0.240	0.201	0.281	0.181	0.270	0.219	0.314	0.237	0.329	0.168	0.272	0.197	0.282	0.247	0.345	0.193	0.308	0.169	0.273	0.201	0.317
	192	0.178	<u>0.268</u>	0.162	0.253	0.201	0.283	0.188	0.274	0.231	0.322	0.236	0.330	0.184	0.289	0.196	0.285	0.257	0.355	0.201	0.315	0.182	0.286	0.222	0.334
	336	0.197	<u>0.289</u>	0.178	0.269	0.215	0.298	0.204	0.293	0.246	0.337	0.249	0.344	0.198	0.300	0.209	0.301	0.269	0.369	0.214	0.329	0.200	0.304	0.231	0.338
	720	0.243	0.326	0.225	0.317	0.257	0.331	0.246	0.324	0.280	0.363	0.284	0.373	0.220	<u>0.32</u>	0.245	0.333	0.299	0.390	0.246	0.355	<u>0.222</u>	0.321	0.254	0.361
	Avg	0.196	<u>0.285</u>	0.178	0.270	0.219	0.298	0.205	0.290	0.244	0.334	0.251	0.344	<u>0.192</u>	0.295	0.212	0.300	0.268	0.365	0.214	0.327	0.193	0.296	0.227	0.338
ETTh1	96	0.386	<u>0.399</u>	0.386	0.405	0.386	0.395	0.414	0.419	0.423	0.448	0.479	0.464	<u>0.384</u>	0.402	0.386	0.400	0.654	0.599	0.376	0.419	0.513	0.491	0.449	0.459
	192	0.426	0.424	0.441	0.436	0.437	0.424	0.460	0.445	0.471	0.474	0.525	0.492	0.436	<u>0.429</u>	0.437	0.432	0.719	0.631	0.420	0.448	0.534	0.504	0.500	0.482
	336	0.480	<u>0.450</u>	0.487	0.458	<u>0.479</u>	0.446	0.501	0.466	0.570	0.546	0.565	0.515	0.491	0.469	0.481	0.459	0.778	0.659	0.459	0.465	0.588	0.535	0.521	0.496
	720	0.484	0.470	0.503	0.491	0.481	0.47	0.500	<u>0.488</u>	0.653	0.621	0.594	0.558	0.521	0.500	0.519	0.516	0.836	0.699	0.506	0.507	0.643	0.616	0.514	0.512
	Avg	<u>0.444</u>	<u>0.435</u>	0.454	0.447	0.446	0.434	0.469	0.454	0.529	0.522	0.541	0.507	0.458	0.450	0.456	0.452	0.747	0.647	0.440	0.460	0.570	0.537	0.496	0.487
ETTh2	96	0.290	<u>0.340</u>	0.297	0.349	0.288	0.338	0.302	0.348	0.745	0.584	0.400	0.440	0.340	0.374	0.333	0.387	0.707	0.621	0.358	0.397	0.476	0.458	0.346	0.388
	192	0.366	<u>0.392</u>	0.380	0.400	<u>0.374</u>	0.390	0.388	0.400	0.877	0.656	0.528	0.509	0.402	0.414	0.477	0.476	0.860	0.689	0.429	0.439	0.512	0.493	0.456	0.452
	336	0.380	0.409	0.428	0.432	<u>0.415</u>	<u>0.426</u>	0.426	0.433	1.043	0.731	0.643	0.571	0.452	0.452	0.594	0.541	1.000	0.744	0.496	0.487	0.552	0.551	0.482	0.486
	720	0.416	0.437	0.427	0.445	<u>0.420</u>	<u>0.440</u>	0.431	0.446	1.104	0.763	0.874	0.679	0.462	0.468	0.831	0.657	1.249	0.838	0.463	0.474	0.562	0.560	0.515	0.511
	Avg	0.363	0.395	0.383	0.407	<u>0.374</u>	<u>0.398</u>	0.387	0.407	0.942	0.684	0.611	0.550	0.414	0.427	0.559	0.515	0.954	0.723	0.437	0.449	0.526	0.516	0.450	0.459
ETTm1	96	0.325	0.360	0.334	0.368	0.355	0.376	<u>0.329</u>	<u>0.367</u>	0.404	0.426	0.364	0.387	0.338	0.375	0.345	0.372	0.418	0.438	0.379	0.419	0.386	0.398	0.505	0.475
	192	0.375	<u>0.386</u>	0.377	0.391	0.391	0.392	0.367	0.385	0.450	0.451	0.398	0.404	<u>0.374</u>	0.387	0.380	0.389	0.439	0.450	0.426	0.441	0.459	0.444	0.553	0.496
	336	0.396	0.405	0.426	0.420	0.424	0.415	<u>0.399</u>	<u>0.410</u>	0.532	0.515	0.428	0.425	0.410	0.411	0.413	0.413	0.490	0.485	0.445	0.459	0.495	0.464	0.621	0.537
	720	0.454	<u>0.442</u>	0.491	0.459	0.487	0.450	0.454	0.439	0.666	0.589	0.487	0.461	0.478	0.450	<u>0.474</u>	0.453	0.595	0.550	0.543	0.490	0.585	0.516	0.671	0.561
	Avg	<u>0.388</u>	0.398	0.407	0.410	0.414	0.407	0.387	<u>0.400</u>	0.513	0.496	0.419	0.419	0.400	0.406	0.403	0.407	0.485	0.481	0.448	0.452	0.481	0.456	0.588	0.517
ETTm2	96	0.175	0.257	0.180	{0.264}	0.182	0.265	0.175	<u>0.259</u>	0.287	0.366	0.207	0.305	0.187	0.267	0.193	0.292	0.286	0.377	0.203	0.287	0.192	0.274	0.255	0.339
	192	0.239	0.299	0.250	0.309	0.246	0.304	<u>0.241</u>	<u>0.302</u>	0.414	0.492	0.290	0.364	0.249	0.309	0.284	0.362	0.399	0.445	0.269	0.328	0.280	0.339	0.281	0.340
	336	0.303	0.342	0.311	0.348	0.307	0.342	<u>0.305</u>	<u>0.343</u>	0.597	0.542	0.377	0.422	0.321	0.351	0.369	0.427	0.637	0.591	0.325	0.366	0.334	0.361	0.339	0.372
	720	0.395	0.394	0.412	0.407	0.407	<u>0.398</u>	<u>0.402</u>	0.400	1.730	1.042	0.558	0.524	0.408	0.403	0.554	0.522	0.960	0.735	0.421	0.415	0.417	0.413	0.433	0.432
	Avg	0.278	0.323	0.288	0.332	0.286	0.327	<u>0.281</u>	<u>0.326</u>	0.757	0.610	0.358	0.404	0.291	0.333	0.350	0.401	0.571	0.537	0.305	0.349	0.306	0.347	0.327	0.371
1 st count	–	15	17	9	16	2	8	5	3	3	0	0	0	1	0	3	0	0	0	4	0	0	0	0	0

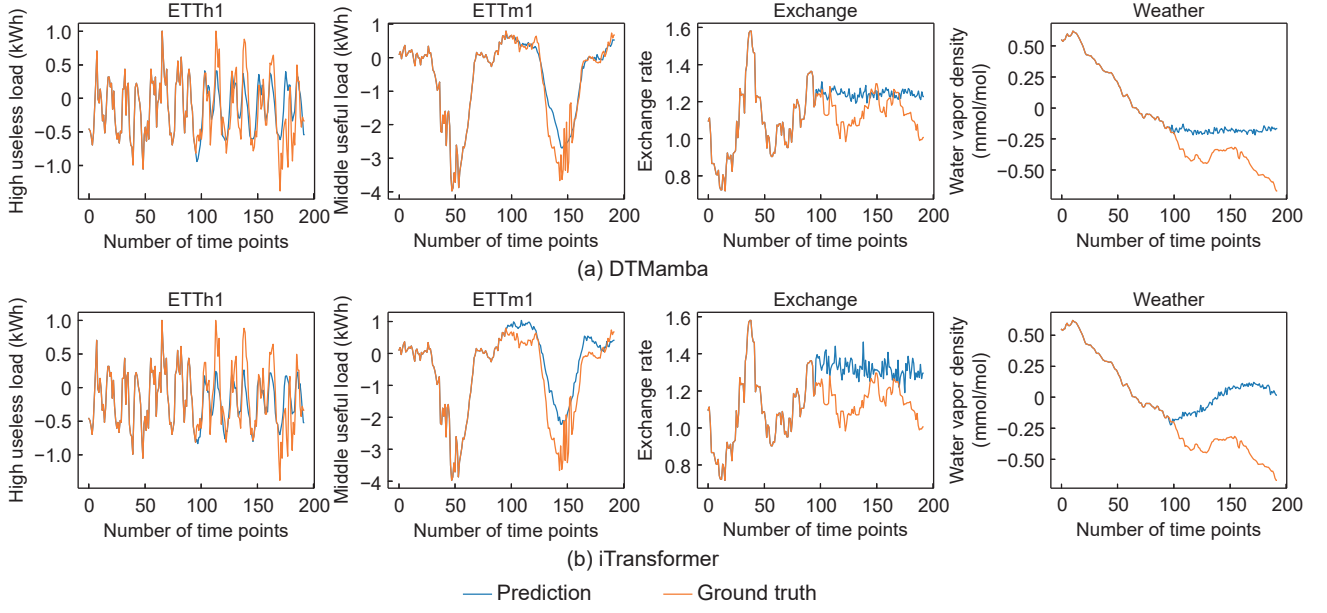


Fig. 2 Visualization of DTMamba and iTransformer on ETTh1/ETTm1/Exchange/Weather when the lookback length $T = 96$ and the forecast length $S = 96$.

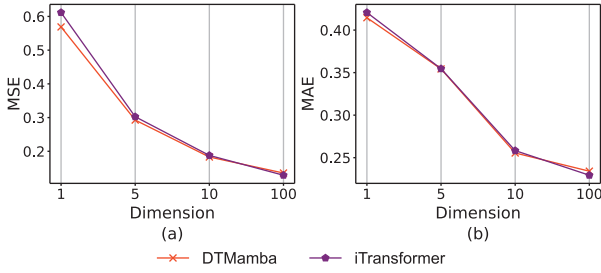


Fig. 3 Average prediction results of DTMamba and iTransformer on Electricity dataset with the forecast length $S \in \{96, 192, 336, 720\}$ (select dimension $n \in \{1, 5, 10, 100\}$).

are also observed in Refs. [11, 12].

4.3 Ablation study and sensitivity test

In this section, we conduct ablation studies on certain modules within the network architecture and test the sensitivity of the model to various hyperparameters in dropout and linear layers and Mamba.

4.3.1 Ablation of residual connections

Ablation studies are conducted on the two residual connections in Section 3.3.2, namely, R^1 and R^2 to verify their effectiveness. Table 5 presents the results on residual connections. We find that the model with

Table 4 Prediction results of DTMamba with lookback length $T \in \{192, 336, 720\}$ and the forecast length $S \in \{96, 192, 336, 720\}$.

Dataset	T	$S = 96$		$S = 192$		$S = 336$		$S = 720$	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	192	0.382	0.398	0.425	0.423	0.461	0.447	0.514	0.5
	336	0.373	0.397	0.411	0.421	0.445	0.445	0.453	0.469
	720	0.386	0.413	0.414	0.431	0.441	0.45	0.463	0.478
ETTh2	192	0.297	0.35	0.369	0.393	0.405	0.426	0.418	0.44
	336	0.289	0.346	0.375	0.401	0.379	0.414	0.412	0.443
	720	0.309	0.357	0.36	0.393	0.371	0.411	0.426	0.453
ETTm1	192	0.296	0.345	0.34	0.374	0.368	0.391	0.436	0.43
	336	0.298	0.344	0.382	0.405	0.428	0.431	0.42	0.423
	720	0.346	0.384	0.35	0.382	0.423	0.431	0.433	0.433
ETTm2	192	0.177	0.258	0.233	0.295	0.3	0.342	0.386	0.393
	336	0.174	0.263	0.235	0.302	0.294	0.337	0.363	0.385
	720	0.176	0.264	0.24	0.314	0.293	0.348	0.371	0.397

Table 5 Residual connection and channel independence ablation experiment with the lookback length $T = 96$; “√” represents the presence the module, while “×” represents the absence of the module.

Residual	Channel independence	Forecast length S	ETTh1		ETTh2		ETTm1		ETTm2	
			MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
√	√	96	0.386	0.399	0.290	0.340	0.325	0.360	0.175	0.257
		192	0.426	0.424	0.366	0.392	0.375	0.386	0.239	0.299
		336	0.480	0.450	0.380	0.409	0.396	0.405	0.303	0.342
		720	0.484	0.470	0.416	0.437	0.454	0.442	0.395	0.394
		Avg	0.444	0.435	0.363	0.395	0.388	0.398	0.278	0.323
×	√	96	0.384	0.408	0.294	0.345	0.329	0.367	0.181	0.263
		192	0.438	0.437	0.373	0.396	0.379	0.393	0.243	0.304
		336	0.482	0.455	0.396	0.418	0.402	0.411	0.307	0.345
		720	0.506	0.485	0.420	0.439	0.467	0.451	0.394	0.395
		Avg	0.453	0.446	0.371	0.400	0.394	0.406	0.281	0.327
√	×	96	0.396	0.408	0.290	0.343	0.376	0.395	0.180	0.262
		192	0.428	0.426	0.387	0.400	0.373	0.413	0.241	0.304
		336	0.487	0.457	0.382	0.409	0.416	0.411	0.302	0.342
		720	0.495	0.482	0.426	0.441	0.543	0.485	0.402	0.399
		Avg	0.452	0.443	0.371	0.398	0.426	0.421	0.281	0.327

residual connections demonstrates a significant improvement in forecasting accuracy.

4.3.2 Ablation of channel independence

Here we perform ablation studies with the channel independence module and the reversed channel independence module. As shown in Table 5, the removal of channel independence means that all dimensions of the time series are considered together, i.e., channel mixing. It can be seen that the model with the channel independence/reversed channel independence modules shows a significant improvement in accuracy.

4.3.3 Ablation of Mamba

To validate the effectiveness of our structure of TMamba blocks (two blocks in serial), we perform Mamba ablation studies on the ETTh1, ETTh2, ETTm1, ETTm2, Weather, and Exchange datasets. As shown in Table 6, “DTMamba” in the “Architecture” column represents the proposed structure, “Mamba” represents the use of a single Mamba model to replace two TMamba blocks, “DMamba” for the use of two TMamba blocks, where each block uses only one Mamba, and “TMamba” for the use of only one TMamba block. From the experimental results, it can be concluded that the proposed DTMamba has the best performance and two sequentially connected TMamba blocks are better at learning relevant features and temporal patterns in time series.

Figure 4 shows the critical difference diagram^[40] for

the Mamba ablation. The methods connected by bold lines have no significant difference in their average ranking. It can be observed that although DTMamba performs best overall, there is no significant difference in the critical difference plot of the average MSE, suggesting that the overall performance might be relatively similar. Similarly, we find that in the critical difference plot of MSE for a prediction length of 720, the advantage of DTMamba is more pronounced than that of the average MSE, suggesting that DTMamba performs better in long-term prediction.

4.3.4 Sensitivity of dropouts

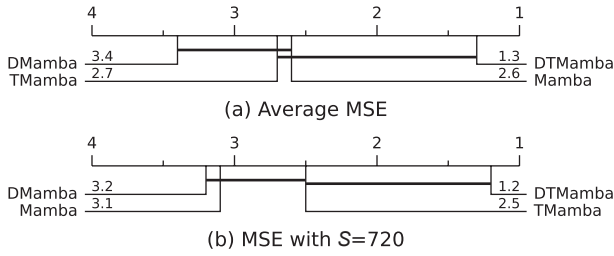
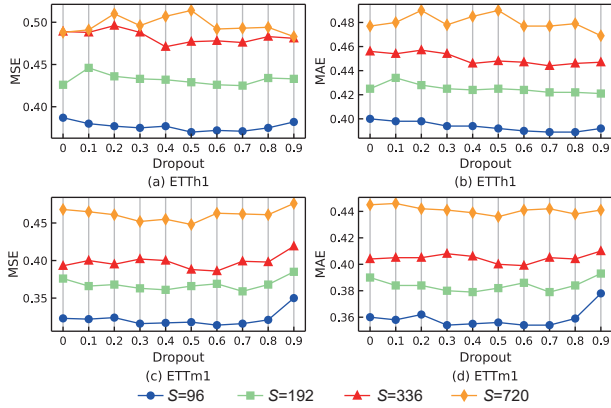
As shown in Fig. 5, except for of the cases with Dropout = 0.2/0.4/0.5 at a prediction length of $S = 720$ in Figs. 5a and 5b and Dropout = 0.9 in Figs. 5c and 5d, no significant fluctuations in Dropout changes overall. Furthermore, the results with Dropout > 0 are in most cases better than Dropout = 0 (i.e., adding Dropout is better). However, increasing the Dropout leads to higher computational costs during training. Therefore, for most experiments, we chose Dropout = 0.05, which is a good balance between effectiveness and computational efficiency.

4.3.5 Sensitivity of linear layer

As mentioned in Section 3.3.1, there are two linear layers in our model, namely Linear_1 and Linear_2. Since the baseline methods, DLinear and RLinear, show the crucial role of linear mapping in previous work on long-term time series prediction, we consider

Table 6 Ablation of Mamba. The best results are in bold.

Architecture	Forecast length S	ETTh1		ETTh2		ETTM1		ETTM2		Weather		Exchange	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
DTMamba	96	0.386	0.399	0.290	0.340	0.325	0.360	0.175	0.257	0.171	0.218	0.083	0.201
	192	0.426	0.424	0.366	0.392	0.375	0.386	0.239	0.299	0.220	0.257	0.173	0.295
	336	0.480	0.450	0.380	0.409	0.396	0.405	0.303	0.342	0.274	0.296	0.346	0.427
	720	0.484	0.470	0.416	0.437	0.454	0.442	0.395	0.394	0.346	0.344	0.868	0.698
	Avg	0.444	0.435	0.363	0.395	0.388	0.398	0.278	0.323	0.253	0.279	0.368	0.405
Mamba	96	0.379	0.398	0.295	0.347	0.321	0.357	0.176	0.259	0.165	0.210	0.085	0.202
	192	0.440	0.426	0.364	0.390	0.373	0.388	0.241	0.301	0.213	0.252	0.172	0.295
	336	0.474	0.444	0.414	0.425	0.398	0.407	0.299	0.339	0.268	0.292	0.333	0.417
	720	0.494	0.479	0.432	0.447	0.462	0.446	0.401	0.397	0.346	0.342	0.928	0.721
	Avg	0.447	0.437	0.376	0.409	0.389	0.400	0.279	0.324	0.248	0.274	0.380	0.409
DMamba	96	0.377	0.396	0.292	0.343	0.326	0.363	0.180	0.258	0.179	0.224	0.086	0.203
	192	0.433	0.426	0.367	0.391	0.373	0.386	0.239	0.299	0.222	0.259	0.175	0.296
	336	0.490	0.456	0.415	0.428	0.442	0.408	0.300	0.338	0.278	0.299	0.345	0.424
	720	0.524	0.492	0.426	0.443	0.460	0.443	0.406	0.400	0.349	0.345	0.901	0.709
	Avg	0.455	0.443	0.375	0.401	0.400	0.400	0.281	0.324	0.257	0.282	0.377	0.408
TMamba	96	0.380	0.396	0.289	0.342	0.318	0.356	0.178	0.259	0.165	0.212	0.084	0.200
	192	0.441	0.430	0.373	0.392	0.369	0.385	0.242	0.303	0.212	0.253	0.176	0.297
	336	0.483	0.449	0.411	0.425	0.405	0.409	0.303	0.341	0.267	0.291	0.347	0.425
	720	0.490	0.475	0.423	0.441	0.463	0.444	0.403	0.400	0.345	0.343	0.925	0.718
	Avg	0.449	0.438	0.374	0.400	0.389	0.399	0.281	0.326	0.247	0.275	0.383	0.410

**Fig. 4** Critical difference diagram on Mamba ablation (numbers in figure represent the ranking).**Fig. 5** Sensitivity of dropouts.

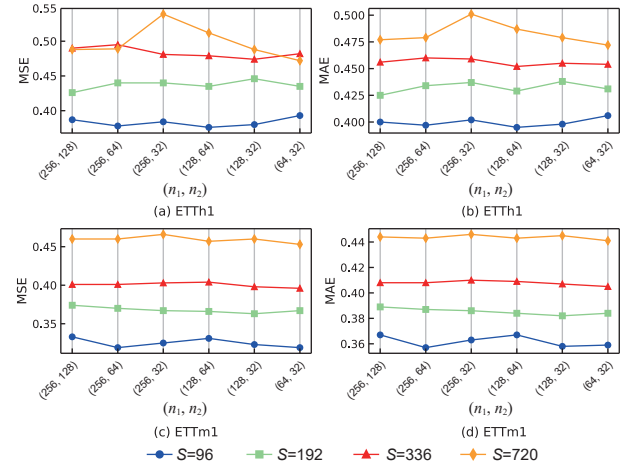
different combinations of linear dimensions to understand the impact of the MLP on the performance

of our network. As shown in Fig. 6, the performance of DTMamba on ETTh1 and ETTm1 under different combinations of linear dimensions does not depend on the choice of dimension combination of two MLPs.

4.3.6 Sensitivity of hyperparameters in mamba

We test the sensitivities of three fundamental hyperparameters in Mamba in the following.

The first factor to consider is the dimension expansion factor (expand). As shown in Fig. 7, an increase in the dimension expansion factor sometimes

**Fig. 6** Sensitivity of linear layer.

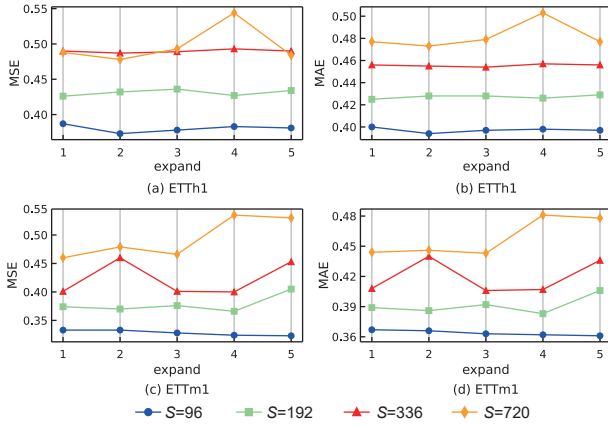


Fig. 7 Sensitivity of dimension expansion factor in Mamba.

leads to a deterioration of the experimental results. Therefore, we chose $\text{expand} = 1$ for all experiments, as this provides better performance.

The second one is the state expansion factor of the SSM (d_{state}) in Mamba. As shown in Fig. 8, except for the case $d_{\text{state}} = 64$ for a prediction length of $S = 720$ in Figs. 8a and 8b, there are no significant fluctuations in the changes of d_{state} . Overall, the results are stable and effective at $d_{\text{state}} = 256$. Therefore, we set $d_{\text{state}} = 256$ for all experiments.

The last one is the dimension of the local convolutional kernel (d_{conv}) in Mamba. As shown in Fig. 9, there are no significant fluctuations in the changes of d_{conv} . Increasing the dimension of the local convolution kernel does not improve the accuracy either. Therefore, we set $d_{\text{conv}} = 2$ for all experiments.

5 Conclusion

In this paper, we propose a long-term time series forecasting model DTMamba based on Mamba.

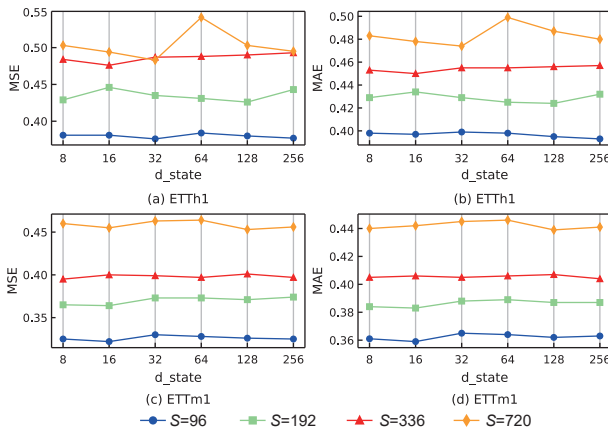


Fig. 8 Sensitivity of state expansion factor in Mamba.

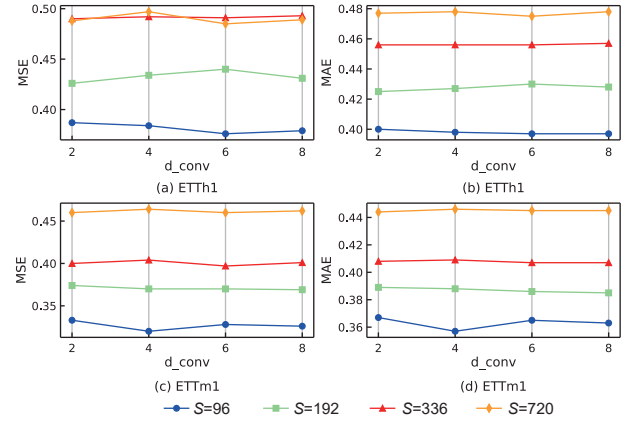


Fig. 9 Sensitivity of local convolutional dimension in Mamba.

DTMamba utilizes channel independence processing to cope with the overfitting problem and takes advantage of the proposed novel TMamba blocks to capture long-term dependencies in time series. Compared to the other baseline models, DTMamba demonstrates better performance. In particular, DTMamba shows clearer advantages with low-dimensional data and the prediction of longer time periods (larger forecast lengths S). In the future, we will further improve DTMamba to fully exploit its performance on long-term and low-dimensional data while improving its performance on short-term and high-dimensional data.

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